

DATA MINING & MACHINE LEARNING

FakeNews Detector

Data Mining & Machine Learning Project

M.Sc. in Artificial Intelligence and Data Engineering — Academic Year 2025–2026

Alessio Grillo — ID: 654947 | Andrea Giacomazzi — ID: 720164

[13/06/2026]

Presentation Outline

01

Problem & Dataset

Task definition · features · EDA

02

Pipeline & Experiments

Preprocessing · model selection · hyperparameter optimization

03

Evaluation & Results

Metrics · cross-validation · final model performance

04

Explainability

Global & local techniques · consistency with domain knowledge

05

Critical Discussion

Error analysis · limitations · future directions

PART 1

PART 2

PART 1 — TECHNICAL

Problem, Pipeline & Results

Problem definition · feature engineering · model selection
hyperparameter optimization · evaluation · test-set results

Problem Definition & Dataset

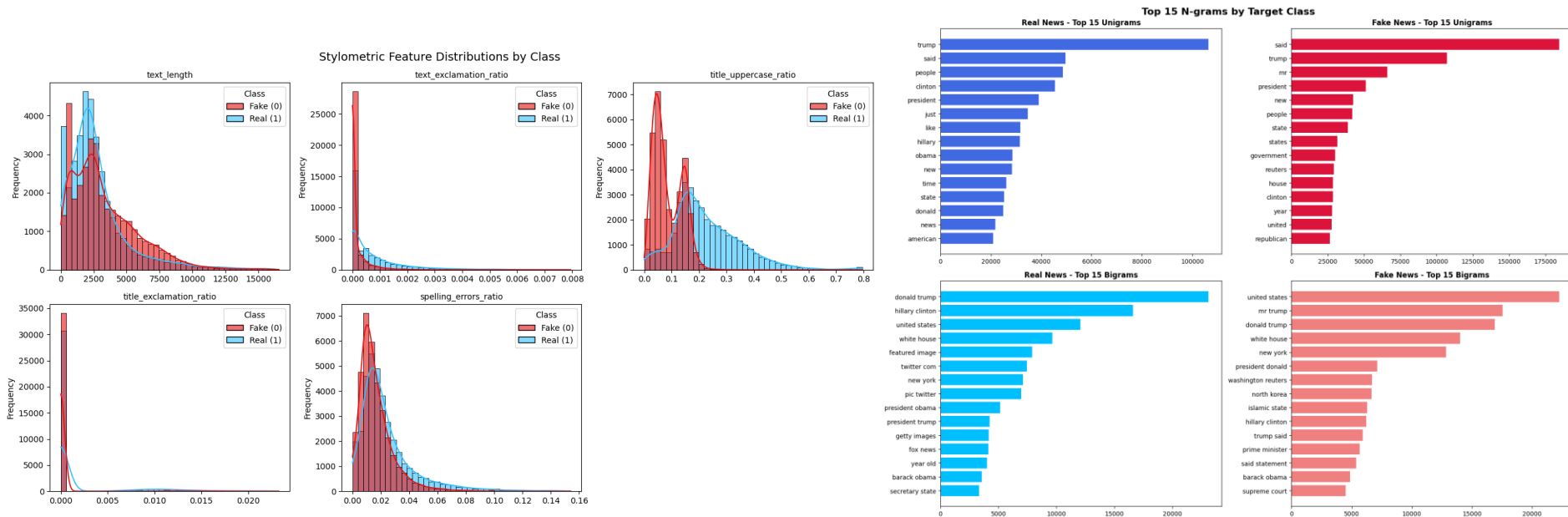
The Problem

- Given a **news article**, predict whether it is credible or deceptive based on its text alone.
- Task type: binary classification
- Fake news spreads faster than anyone can fact-check it, shaping opinions on elections, health, and public trust.
- A model cannot check whether something is true facts, it can only detect stylistic and linguistic patterns. And the best fake articles are written to sound exactly like real ones.

Dataset at a Glance

Task	Binary classification
Dataset	Kaggle, 72.134 samples, 3 features
Target	Article veracity (0: Fake News, 1: Real News)
Features	1 numerical, 2 textual
Missing values	0,21% overall
Class balance	51.5% / 48.5% balanced
Split	80% Training / 20% Testing
Metric(s)	Accuracy, F1-score, Precision, Recall, AUC-ROC

Exploratory Data Analysis



Key findings from EDA:

Stylometric features reveal significant structural and stylistic differences between classes, proving that how a news article is written is highly indicative of its veracity.

N-gram The N-gram analysis reveals strong dataset-specific biases rather than purely semantic differences.

Feature Engineering & Selection

Engineering

- Text vectorization using TF IDF
- 10,000 TF IDF features from concatenated title + body.
- Extraction of 5 stylometric features: title uppercase ratio, text length, spelling error rate, exclamation & question mark density (both title and text).
- MinMax scaling on the 5 stylometric features to align their magnitude with the sparse TF IDF matrix.
- Concatenated the sparse TF IDF matrix with the dense, scaled stylometric features to create a unified feature space.

Selection Method

- Frequency based Filtering: Limited the TF IDF vocabulary to the top 10,000 most frequent tokens to reduce dimensionality and eliminate rare noise/typsos (we discarded some stopwords based on the n gram plot.
- Evaluated the unified feature pool using a Chi Square test.
- Title uppercase ratio emerged as the first overall predictor, and title exclamation ratio ranked in the Top 20.
- Retained features with lower Chi Square scores (text length, exclamation ratio, spelling errors) because EDA shows they contain critical structural anomalies. This preserves domain robustness against keyword overfitting.

Why These Features?

- TF-IDF captures vocabulary shifts and source-mimicking artifacts confirmed by EDA.
- Stylometric features target sensationalism signals that differ significantly between classes.

ML Pipeline: Avoiding Data Leakage & Bias



Data Leakage Prevention

- All preprocessing steps (TF-IDF, MinMax scaler, stopwords removal) wrapped in a Scikit-Learn Pipeline, fitted exclusively on training folds.
- Train/test split performed before any transformation. Test folds only ever see frozen parameters.
- No test set distribution ever influences the vocabulary, scaling bounds, or stopwords list.

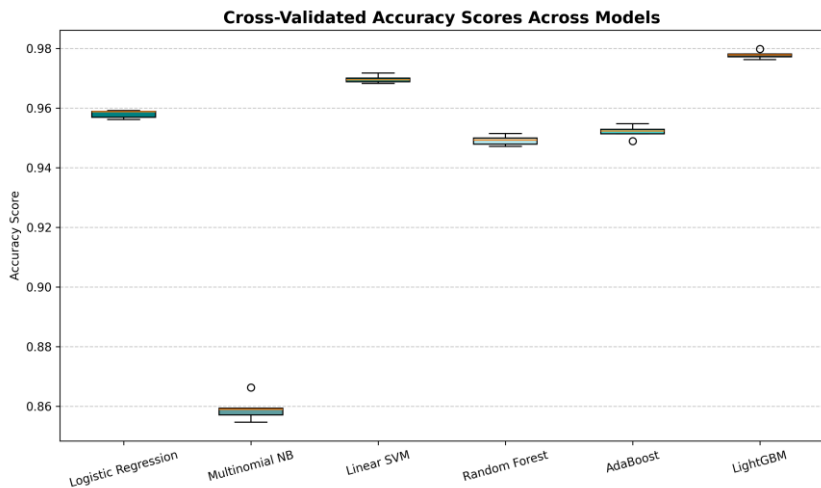
Class Imbalance Handling

- No balancing technique applied. The dataset is naturally balanced (51.5% fake / 48.5% real), so SMOTE and class weights were unnecessary.
- Stratified 5-fold CV preserves the natural class ratio across every split, ensuring stable and comparable metrics throughout.

Metric Justification

- **Primary: F1-score & Accuracy.** Balanced classes make accuracy meaningful; F1 jointly captures precision and recall.
- AUC-ROC reported as secondary metric to assess discriminative power across all decision thresholds.

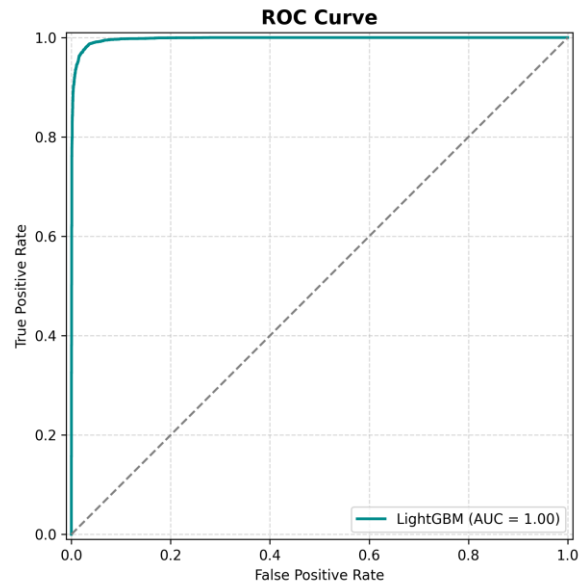
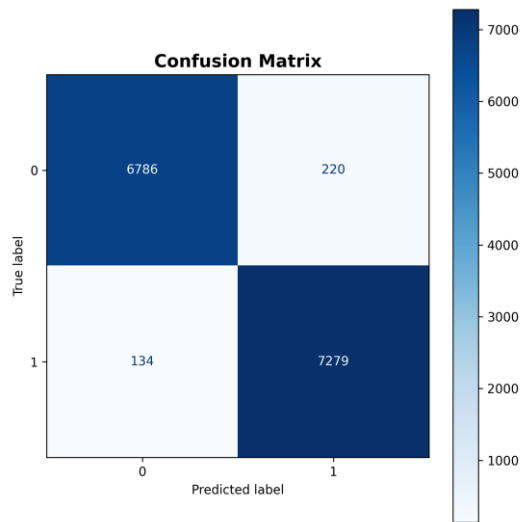
Model Selection & Hyperparameter Optimization



Key Decisions

Candidate models	Linear baseline (LR, NB, SVM) + ensemble (RF, AdaBoost, LightGBM) covering linear, probabilistic, geometric and boosting families
CV strategy	Stratified 5-fold CV for all models
Selection criterion	Accuracy
Chosen model	LightGBM: highest CV accuracy (0.978) with lowest variance
Search strategy	Random Search (5 combinations x model)
Search space	$n_estimators$: {50,100,150} · $learning_rate$: {0.05, 0.1} · num_leaves : {15, 31}
Best config	$n_estimators$: 100 · $learning_rate$: 0.1 · num_leaves : 31
Statistical test	McNemar's test: LightGBM vs Linear SVM · $\chi^2=20.32$ · $p=0.00001$, $\alpha=0.05$

Test-Set Results



Key Metrics (Test Set)

97,5%

Accuracy

97,6%

F1-Score

99,7%

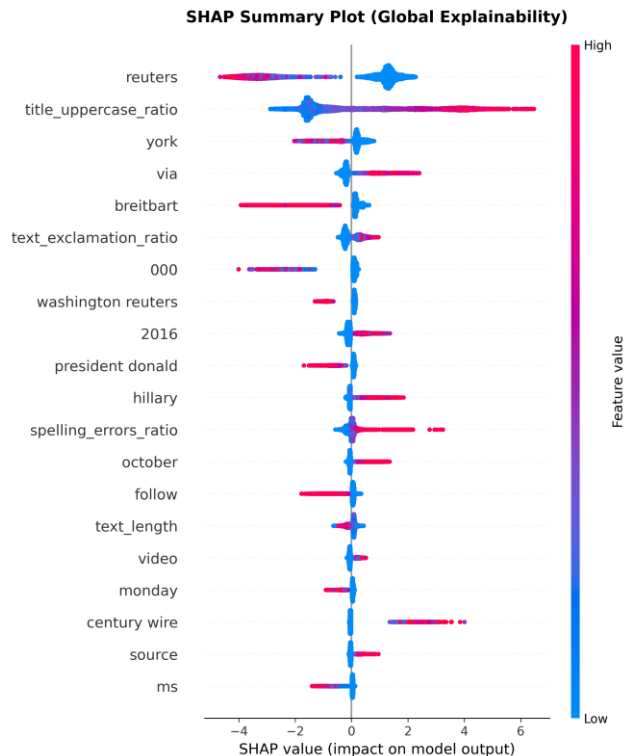
AUC-ROC

PART 2 — CRITICAL ANALYSIS

Explainability & Critical Discussion

How does the model work? · What did it get wrong and why?
Limitations · future directions · domain impact

Global Explainability: What Drives the Model Overall?

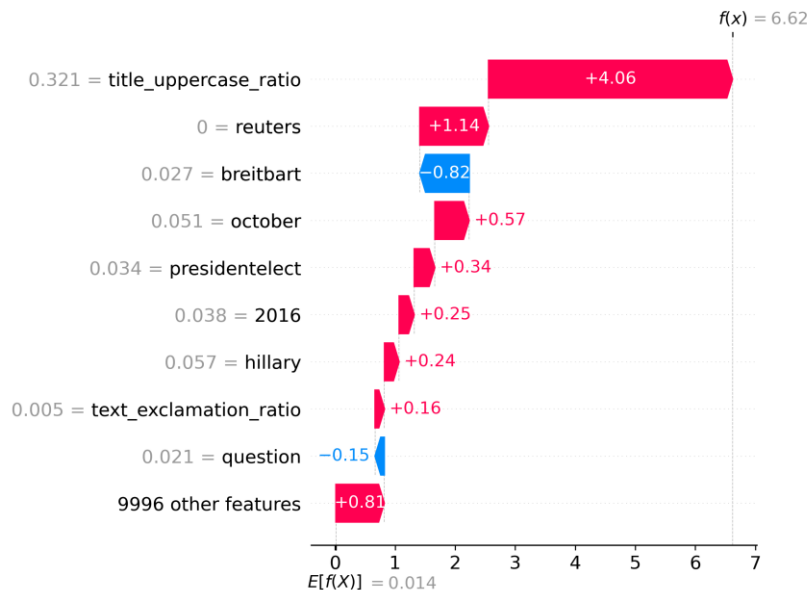


Interpretation

- **SHAP beeswarm analysis:**
 - Each dot = one sample
 - Color = feature value
 - X-axis = feature impact
- **Key signals:**
 - Presence of *Breitbart* -> pushes prediction toward Fake
 - Absence of *Reuters* -> associated with Real
- **Insight:**
 - Model learns **journalistically** meaningful signals, not dataset artifacts.

Local Explainability: Why This Specific Prediction?

SHAP Waterfall Plot (Local Explainability - Index 2)

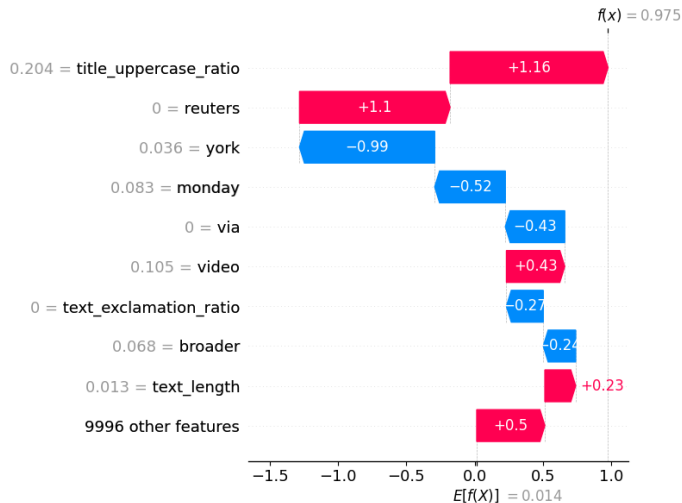


Case Analysis

- **Sample:** Article from October 2016 mentioning "president-elect", "Hillary", and "2016" — political content from the US election period, ground truth: Real News (correctly classified)
- **Key drivers:** `title_uppercase_ratio` (+4.06) dominates; `reuters` absent (value=0, +1.14) means the absence of this token pushes toward real; `breitbart` presence drives toward fake as expected
- **Local & Global alignment:** top features here (`title_uppercase_ratio`, `reuters`, `breitbart`) match the global SHAP ranking — consistent behavior across instances.
- **Conclusion:** An article with high capitalization, no press agency signature, and partisan political references make the model reasons correctly.

Error Analysis: What Did the Model Get Wrong and Why?

SHAP Waterfall Plot (Local Explainability - Index 61)



Reverse-Engineering Errors

- **Two failure modes:** FN (134): fake news mimicking legitimate reporting; FP (220): genuine opinion content penalized for aggressive style
- **Root cause:** the model confounds writing style with factual content, stylistic proxies cannot capture semantic intent
- Index 61 (FP): *title_uppercase_ratio* and *reuters* absent push toward Real, but *york* and *monday* pull back.
- Errors are **systematic**, not random: misclassifications cluster around the stylistic decision boundary, confirming a known architectural limit of TF-IDF pipelines

Comparison with Prior Work

Compared to Tian et al. (2025), which uses TF-IDF for ML and learned token representations for DL models, our LightGBM Pipeline achieves **97.5%** accuracy with TF-IDF + 5 stylistic features, while remaining **fully interpretable** via SHAP.

Domain Impact, Limitations & Future Directions

Domain Impact

- **First-pass filter** for content moderation, flags suspicious articles before publication or amplification.
- Supports fact-checkers by prioritizing review queues, not replacing human judgment.
- Balanced error rate (FP censors journalism; FN lets misinformation spread) makes it a decision-support tool, not a final arbiter.
- SHAP explanations make predictions auditable, essential for human-facing moderation.

Limitations

- WELFake is **English-only** and **US-politically** skewed, generalization untested.
- Static TF-IDF is vulnerable to **concept drift**; performance degrades without retraining.
- The model detects stylistic proxies of deception, not factual truth.
- Evaluated on a single dataset split, no real-world feedback loop.

Future Directions

- Replace TF-IDF with contextual embeddings (**BERT**) for deeper semantic understanding.
- Integrate **fact-checking APIs** to cross-reference claims against verified databases.
- Extend to multimodal input (images + metadata) to catch deception beyond text.
- Deploy with **human feedback** to measure calibration and detect drift.

Thank You

Conclusions:

- Problem: automated binary classification of news articles to combat digital misinformation, using semantic and stylometric features
- Best model: LightGBM with 97.5% accuracy and F1, validated via Stratified 5-Fold CV and confirmed statistically superior to Linear SVM (McNemar's, $p=0.00001$)
- Key XAI finding: SHAP reveals the model learned journalistically meaningful signals, source tokens (breitbart, reuters) and sensationalist stylometry; not dataset artifacts, confirming genuine generalization
- Main limitation & next step: TF-IDF cannot distinguish writing style from factual content; future work should explore transformer-based architectures (BERT, RoBERTa) for deep contextual understanding

Alessio Grillo · a.grillo6@studenti.unipi.it

Andrea Giacomazzi · a.giacomazzi1@studenti.unipi.it